



Research Report

SICHEN SHAWN CHAO

China's Retirement Reform and the 2035 Statutory Working-Age Window

Buying Time amid Demographic Decline and AI-Driven Change

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About This Report

China's 2025 retirement reform raised some of the world's lowest statutory retirement ages as the country's working-age population entered sustained decline and as artificial intelligence began to reshape its labor market. This Research Report uses a monthly birth-cohort simulation to estimate the statutory working-age window created by the reform and examines its implications for China's 2035 modernization agenda, productivity, pension timing, and labor-market adjustment.

This analysis should interest policymakers and analysts concerned with China's economic trajectory, demographic and pension policy, and the labor-market effects of AI. It is part of a broader RAND China Research Center project on China 2035.

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Summary

China's 2025 retirement reform creates a temporary statutory working-age window rather than a solution to population aging. Using a monthly birth-cohort simulation, this report estimates the reform's effect through 2050 and examines what would be required for the additional statutory headcount to become productive economic capacity. In the preferred scenario, the reform keeps the statutory working-age population broadly stable through the mid-2030s and leaves about 48 million more people below statutory retirement thresholds in 2035 than under the old rules. Its direct contribution to growth is limited, however, and its broader value depends on older-worker participation, employer demand, productivity, service-sector absorption, and how artificial intelligence affects the occupations represented among the retained cohorts.

- The reform creates a temporary statutory working-age window, not a solution to aging. In the preferred central scenario, it holds China's statutory working-age population roughly flat through the mid-2030s—a rise of about 5.5 million between 2024 and 2035, against a decline of roughly 43 million under the old rules. Relative to the old rules, it keeps about 28 million additional people below the statutory retirement threshold by 2030 and about 48 million by 2035.
- The period of broad stability ends after the mid-2030s; the reform effect itself remains large into the 2040s. The reform delays the decline in the working-age population but does not reverse it. The magnitude of the buffering effect peaks in the early-to-mid 2040s at roughly 66 to 79 million additional people depending on specification, and then it erodes as the statutory population resumes a sustained decline. This quantity refers to statutory headcount, not realized employment.
- The cushion depends disproportionately on extending the working lives of blue-collar women. Women account for roughly 54 percent of the 2035 effect. Blue-collar women on the lower-wage “old-50” track, who receive the reform's largest increase of five years, supply about 38 percent of the total.
- The reform's direct contribution to growth is small because labor had already stopped driving Chinese economic growth. Over 2016–2023, labor contributed essentially nothing to GDP growth while capital and productivity carried almost all of it. Even on the generous assumption that every retained person works, the reform would leave annual growth through 2035 only about 0.3 percentage points higher than it would be under the old rules. This is a minuscule number relative to the roughly 4.17 percent average annual GDP growth target that Beijing has identified as necessary to achieve its 2035 income target.
- Whether the window becomes true economic capacity depends on productivity, demand, and AI's effect on the retained workers. Meeting the 2035 target increasingly falls on productivity as capital's contribution fades. Many women assigned to the old-50 proxy are in frontline, production, service, or manual occupation groups. Whether AI complements or substitutes for tasks prevalent in those occupations is an open empirical

question that this model does not estimate; the report quantifies the window itself, not what AI does with it. The reform is therefore a complement to, but not a substitute for, China's productivity agenda. Its payoff will depend on whether older-worker participation, employment, and labor-intensive service sectors actually absorb the retained cohorts.

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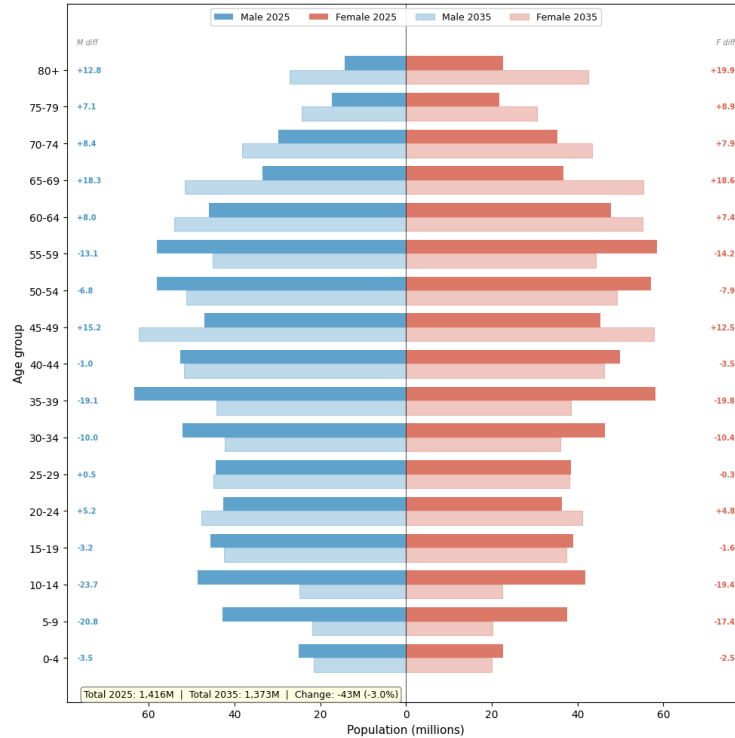
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Chapter 1. Main Claim: A Window, Not a Solution

China's pool of working-age people is shrinking. That decline weighs directly on economic growth, and the impact deepens over time. Beijing has been searching for policy levers that can slow the loss. China's January 2025 retirement reform creates a temporary statutory working-age (SWA) window through 2035. It buys time, but it does not solve population aging by itself.

China's historic population decline reflects both past policies and current economic constraints. Decades of birth restrictions, including the one-child policy, reduced the size of younger cohorts and reshaped family norms; urbanization and the rising cost of raising children have further discouraged births even after birth limits were relaxed ("China's Population Drops for Fourth Year as Fewer Babies Born," 2026). China has one of the world's highest childrearing costs relative to gross domestic product (GDP) per capita ("China's Childcare Costs Among Highest in World—Think Tank," 2024). In 2025, births fell to 7.92 million while deaths reached 11.31 million, marking China's fourth consecutive year of population decline ("China's Population Drops for Fourth Year as Fewer Babies Born," 2026). These conditions have produced an increasingly top-heavy population pyramid, with smaller cohorts of future labor-market entrants and growing older age groups (Figure 1.1). Fewer young cohorts will enter the labor market, and larger older cohorts place rising pressure on pensions, health care, and elder care.

Figure 1.1. China's Population Pyramid, 2025 and 2035



SOURCE: Author's calculations using United Nations, Department of Economic and Social Affairs, Population Division, 2024.

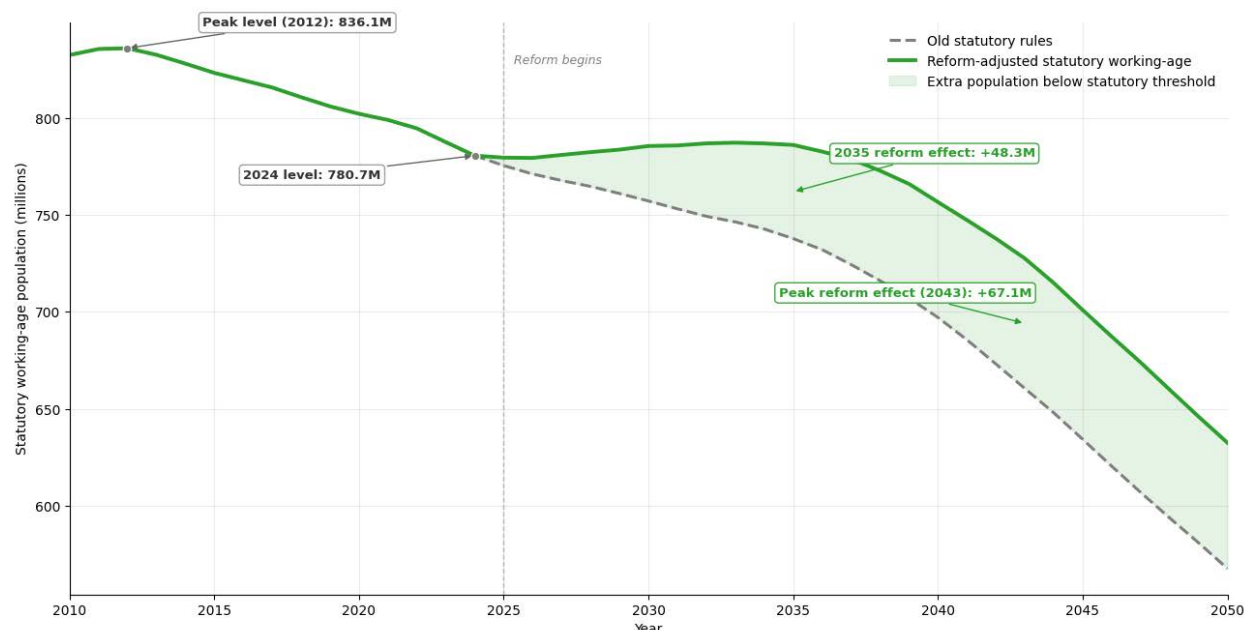
NOTE: Population values are shown in millions. Projections for 2025 and 2035 use the United Nations (UN) World Population Prospects (WPP) 2024 medium variant.

In response to these pressures, China adopted a retirement reform in September 2024 that took effect in January 2025, raising some of the world's lowest statutory retirement ages. Over a nearly 15-year phase-in, men's retirement age rises from 60 to 63, women in white-collar tracks from 55 to 58, and women in blue-collar tracks from 50 to 55 (Standing Committee of the National People's Congress, 2024; "China Approves Plan to Raise Retirement Age from January 2025," 2024).¹ A monthly birth-cohort simulation of the reform's effects on the SWA population shows that these increases create a potential buffer for China's aging population. By keeping tens of millions of additional people below the statutory retirement threshold through roughly 2035, the policy buys Beijing time (Figure 1.2). The key question is whether China can convert that

¹ The reform is phased in by birth cohort, not through a single calendar-year jump in retirement age. For men and women in the old 55-year track, the statutory retirement age is delayed by one month for every four months of birth cohort, capped at a 36-month increase; for women in the old 50-year track, it is delayed by one month for every two months of birth cohort, capped at a 60-month increase. Although the birth-cohort ramp reaches the full new thresholds over roughly 10–12 birth years depending on track, the affected cohorts reach retirement between 2025 and 2039, so the reform is commonly described as a roughly 15-year phase-in.

time into real productivity gains by capitalizing on longer effective working lives, artificial intelligence (AI) diffusion, automation, and service-sector expansion.

Figure 1.2. China’s Retirement Reform Creates a Temporary SWA Window



SOURCE: Author’s calculations using data from United Nations, Department of Economic and Social Affairs, Population Division, 2024; United Nations Statistics Division, 2026; and the reform schedule in Standing Committee of the National People’s Congress, 2024.

NOTE: Preferred central scenario, S5 (cohort-trended occupation model, $\theta = 0.25$, $\beta = 0.5$). θ is the share of women in service and sales occupations assigned to the white-collar track, varied across scenarios because no Chinese dataset reports this split directly. β sets how much of women’s movement from blue-collar into white-collar work, observed within birth cohorts between the 2010 and 2020 censuses, is projected forward after 2020. Full definitions appear in Appendix F. The shaded area shows the additional population kept below the statutory retirement threshold under the 2025 reform relative to the old rules. This is a statutory headcount, not realized employment or labor-force participation.

That window coincides with a formal planning horizon. The 14th Five-Year Plan Outline and Long-Range Objectives Through 2035 frame 2035 as the year by which China aims to basically achieve socialist modernization (“Outline of the 14th Five-Year Plan for National Economic and Social Development and Long-Range Objectives Through the Year 2035 of the People’s Republic of China,” 2021). The 2025 AI Plus guidance uses the same horizon, calling for China to enter a new stage of “intelligent economy and intelligent society” by 2035 (State Council of the People’s Republic of China, 2025). Although the simulation identifies a statutory window that coincides with China’s planning timelines, it cannot directly establish policymaker intent. The retirement reform’s public rationale centers on demographic change, human-resource utilization, and pension timing, not technology policy (Standing Committee of the National People’s Congress, 2024; “China Approves Plan to Raise Retirement Age from January 2025,”

2024). The reform creates a SWA and pension-timing window that coincides with Beijing's 2035 modernization horizon.

Put simply, it is not just how many people remain legally below retirement thresholds, but also whether those additional people remain employed, continue contributing, are demanded by firms, and can be made more productive through AI-enabled augmentation, service-sector growth, retraining, and longer effective working lives.

Chapter 2. Why Statutory Working-Age Population Matters

Conventional working-age measures, such as the 15–64 band used in United Nations demographic tabulations (United Nations, Department of Economic and Social Affairs, Population Division, 2024) and China’s official 16–59 band (National Bureau of Statistics of China, 2025b), are useful for broad demographic comparison. However, in the context of China, they miss a central institutional feature of its retirement system: statutory retirement rules historically differed by gender and occupation. Under the pre-reform system (from 1978 until January 2025), male employees (男职工) generally retired at 60, women in cadre, managerial, or professional-technical posts (女干部 / 管理技术岗位女职工) at 55, and women classified as workers—often in blue-collar, frontline, or production roles (女工人 / 生产操作岗位女职工)—at 50 (State Council of the People’s Republic of China, 1978; Standing Committee of the National People’s Congress, 2024). The SWA measure is therefore a sharper proxy for the population legally below retirement thresholds than a flat age-band measure, and it allows the reform’s mechanical effect to be quantified as the increases phase in. For reference, China’s National Bureau of Statistics (NBS) reported 858.0 million people aged 16–59 at year-end 2024, while the constructed statutory measure in the preferred central scenario based on the three retirement cutoffs was calculated to be roughly 781 million (National Bureau of Statistics of China, 2025b). The discrepancy reflects women aged 50–59 (old-50 track) and 55–59 (old-55 track) who are past their statutory thresholds but inside the 16–59 band; men do not contribute, because their threshold coincides with the band’s edge.

At the same time, the SWA population should not be confused with actual labor supply. Some people below the statutory retirement age are out of the labor force; some above it continue working, especially in informal or agricultural work. Many older rural migrants and retirees continue working because pension benefits are inadequate, while the 2024 retirement reform itself permits flexible early or delayed retirement under specified conditions (“In Rapidly Ageing China, Millions Can’t Afford to Retire,” 2024; Standing Committee of the National People’s Congress, 2024). Employer demand, health, disability, early-retirement choices, and pension eligibility all affect realized employment. The value of the model is that it isolates the policy lever: the change in legal thresholds and the population affected by those thresholds.

Chapter 3. Data and Methods

This report estimates how China’s January 2025 statutory retirement reform changes the country’s statutory working-age population through 2050. The object of analysis is the number of people below the statutory retirement age in each year. That is the quantity the reform directly moves. The model counts the population from age 16, matching the lower bound of China’s official 16–59 working-age population category, up to each group’s applicable statutory retirement age (National Bureau of Statistics of China, 2025b). The headcount is distinct from realized labor supply: it does not predict how many people will work, retire, or participate in the labor force. Chapter 6 returns to that distinction.

The estimate comes from a monthly birth-cohort simulation, detailed in Appendix G. For each year, the model divides the population by sex and single year of age into monthly subcohorts and evaluates whether each subcohort remains below the applicable statutory retirement threshold at the model’s mid-year reference point. Each subcohort is assigned a statutory retirement age under two rule sets: the pre-reform rules in effect through 2024 and the reform’s phased schedule beginning in January 2025. The reform raises retirement ages in monthly increments, subject to the statutory three- and five-year caps. The reform effect is the difference between the computed SWA populations under the old rules and the reform. Because both paths are built on the same population data, the difference isolates the mechanical effect of the policy change.

The analysis draws on four main data sources, detailed in Appendix A. Population by sex and single year of age comes from the UN World Population Prospects 2024, accessed through the UN Population Data Portal, with estimates through 2023 and projections from 2024 onwards (United Nations, Department of Economic and Social Affairs, Population Division, 2024). Female retirement-track shares are constructed from UN-published Chinese census occupation tabulations for 2010 and 2020 under the International Standard Classification of Occupations, 2008 revision (ISCO-08), by sex and five-year age group (United Nations Statistics Division, 2026). Historical urbanization rates from 1949 onward are drawn from China’s National Bureau of Statistics and are used only in the legacy urbanization-based specification described below (National Bureau of Statistics of China, 2025a). The growth-accounting context in Chapter 7 uses Penn World Table 11.0 (Feenstra, Inklaar, and Timmer, 2015).

The central modeling choice is how to divide women between the lower, old-50 track and the higher, old-55 track, because the reform treats the two tracks differently. Women in the old-50 track move from retirement at 50 to retirement at 55, a five-year increase. Women in the old-55 track move from 55 to 58, a three-year increase. Assigning a large proportion of women to the old-50 track would increase the estimated reform effect, because more women receive a larger statutory increase. This split cannot be clearly determined from official data. A woman’s track

follows her cadre or worker job classification, which China does not publish as a population aggregate, so no comprehensive public dataset reports how the female working-age population is divided between the two retirement categories. The share within each track must therefore be estimated rather than observed.

The analysis reports seven specifications built on two approaches. The first is a legacy urbanization-based specification, S1, detailed in Appendix E. It uses the urban share of the population as a proxy for the share of women on the white-collar, old-55 track. This was a simple starting proxy, but it has a clear weakness. The urban share overstates the white-collar female share: in the 2020 census year, it implies that roughly 64 percent of women are on the higher track, compared with roughly 22 percent in the census occupation data from the UN (United Nations Statistics Division, 2026). Because the old-55 female group receives the smaller statutory increase, this overstatement of its proportion tends to understate the reform effect. S1 is therefore retained as a legacy high-old-55 comparison and lower-effect benchmark, not as the preferred estimate or a formal lower bound.

The second approach is occupation-grounded. Specifications S2 through S7 assign women to retirement-track proxies using census occupation shares under ISCO-08 rather than using urbanization. These specifications vary two assumptions: how service-and-sales occupations are allocated between the tracks, and how far occupation shares are projected across birth cohorts. The central specification is S5, which sets $\theta = 0.25$ and $\beta = 0.5$. Here, θ is the share of service-and-sales workers assigned to the old-55 track, and β governs how much of the observed 2010–2020 occupational upgrading is projected forward. The occupation-grounded model is the primary basis for the report’s estimates. Full definitions of the seven specifications appear in Appendix F.

The simulation went through several checks to verify the accuracy of the input data, the implementation of the policy rules, and the sensitivity of the results to uncertainty. Model inputs and policy parameters were checked against their sources. The retirement-schedule implementation was checked at multiple checkpoints for every cohort’s timeline: initial policy phase-in, intermediate phase-in points, and the concluding statutory three- and five-year caps. The exact monthly simulation was cross-validated against an approximate cutoff method, which matched the old-rules series and converged with the exact method once the reform fully phased in. The seven specifications serve as sensitivity tests around the female retirement-track assumptions. Key numerical outputs were checked manually and programmatically. The coding notebook runs from a clean state with all checks passing.

Chapter 4. Simulation Results

The monthly cohort simulation applies the official reform schedule by birth month rather than approximating it as a flat age increase. Men and women in the old 55-year track gain one retirement-age month for every four birth months, capped at 36 months; women in the old 50-year track gain one month for every two birth months, capped at 60 months (Standing Committee of the National People’s Congress, 2024). The corresponding pre-reform SWA population is computed in the same framework using the old thresholds, so the two series are produced from a common population basis and differ only in the retirement age applied.

In the preferred occupation-grounded central scenario (S5), China’s SWA population stood at roughly 781 million in 2024, the last full year before the reform. Compared with a steep decline in the SWA population under the old rules, the reform-adjusted statutory working-age population remains broadly stable through the early-to-mid 2030s. In S5, the SWA population rises by 5.5 million between 2024 and 2035, compared with a 42.8 million decline under the old-rules counterfactual. Across the scenarios, which vary the female split, the 2024–2035 change ranges from a decline of 7.9 million to an increase of 6.8 million. Compared with a future under the old rules, S5 retains 28 million additional people below the statutory threshold by 2030, about 48 million by 2035, and roughly 67 million at peak in 2043. Across all scenarios, the 2035 effect ranges from about 45 million to 55 million additional people (Table 4.1).

Table 4.1. Reform Effect by Scenario

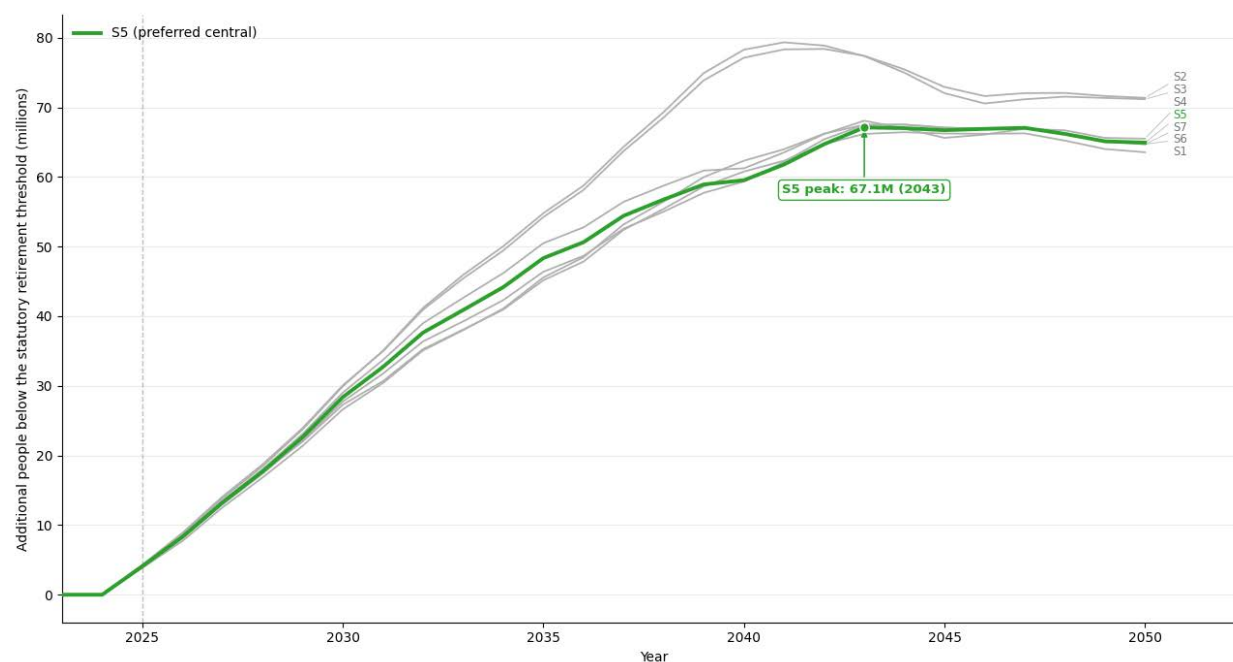
Code	Scenario	θ	β	2030	2035	2040	Peak
S1	Legacy urbanization proxy (retained for comparison)	—	—	+27.3	+45.2	+60.8	+66.4
S2	Static 2020 occupation shares, all-age	0	0	+30.1	+54.2	+77.1	+78.4
S3	Static 2020 occupation shares, age-specific	0	0	+29.9	+54.8	+78.3	+79.3
S4	Cohort-trended occupation model	0	0.5	+29.0	+50.5	+61.3	+68.1
S5	Cohort-trended occupation model (preferred central)	0.25	0.5	+28.4	+48.3	+59.5	+67.1
S6	Cohort-trended occupation model	0.5	0.5	+27.8	+46.4	+59.4	+67.5
S7	Cohort-trended occupation model	1	0.5	+26.6	+45.6	+62.3	+67.5

SOURCE: Author's calculations using data from United Nations, Department of Economic and Social Affairs, Population Division, 2024; United Nations Statistics Division, 2026; National Bureau of Statistics of China, 2025a; and the reform schedule in Standing Committee of the National People's Congress, 2024.

NOTE: Values are additional people below the statutory retirement threshold under the reform relative to the old-rules counterfactual, in millions, at the indicated year (Peak = each scenario's maximum). θ and β are sensitivity parameters governing the female retirement-track split and the projection of occupational change between the 2010 and 2020 censuses; both are defined in Appendix F. The legacy proxy (S1) predates the occupation-based framework and takes neither parameter. These are statutory headcounts, not realized employment or labor-force participation.

However, even with the reform, the SWA population resumes a sustained decline in the late 2030s. The number of additional people retained below the statutory retirement threshold peaks in the early-to-mid 2040s at roughly 66 million to 79 million, depending on specification. The reform delays the decline in the SWA population; it does not eliminate it (Figure 4.1).

Figure 4.1. The Reform’s Buffering Effect Rises, Peaks in the Early-to-Mid 2040s, and Then Erodes



SOURCE: Author’s calculations using data from United Nations, Department of Economic and Social Affairs, Population Division, 2024; United Nations Statistics Division, 2026; National Bureau of Statistics of China, 2025a; and the reform schedule in Standing Committee of the National People’s Congress, 2024.

NOTE: Lines show additional people below the statutory retirement threshold under the 2025 reform relative to the old-rules counterfactual, in millions, for the seven scenarios in Table 4.1. The preferred central scenario (S5: cohort-trended occupation model, $\theta = 0.25$, $\beta = 0.5$) is highlighted in green; the two static-share scenarios (S2 and S3) form the upper pair; the remaining scenarios are shown in gray, with exact values in Table 4.1. S2 applies the single all-age 2020 share to every cohort. S3 preserves the 2020 age-specific profile, in which white-collar shares fall with age, so cohorts reaching the threshold carry the lowest old-55 shares. Scenario peaks occur between 2041 and 2044. These are statutory headcounts, not realized employment or labor-force participation.

The scenarios in Table 4.1 are alternative assumptions about the female retirement-track split, which is the model’s largest source of uncertainty that is further examined in Chapter 5. They are not definitive upper and lower bounds on the reform’s possible effect; they hold the rest of the model fixed. The static age-specific specification (S3) produces the largest effect within this family because threshold-age cohorts carry the lowest old-55 shares under its frozen age profile. The legacy urbanization proxy (S1) produces one of the smaller near-term effects. The core finding is robust across the scenario family: the reform creates a large SWA cushion through the mid-2030s, but the cushion remains a legal and demographic measure, not an estimate of realized employment. Chapter 5 explains why the female old-55 versus old-50 split matters for interpreting the scenario range.

The measurements above reflect the number of people below the statutory retirement threshold — not workers and not realized labor supply. Converting the SWA population into employment, output, and pension contributions depends on labor-force participation, employer demand, informality, and effective retirement behavior, none of which the simulation estimates.

Chapter 5. Why the Female 50/55 Split Matters

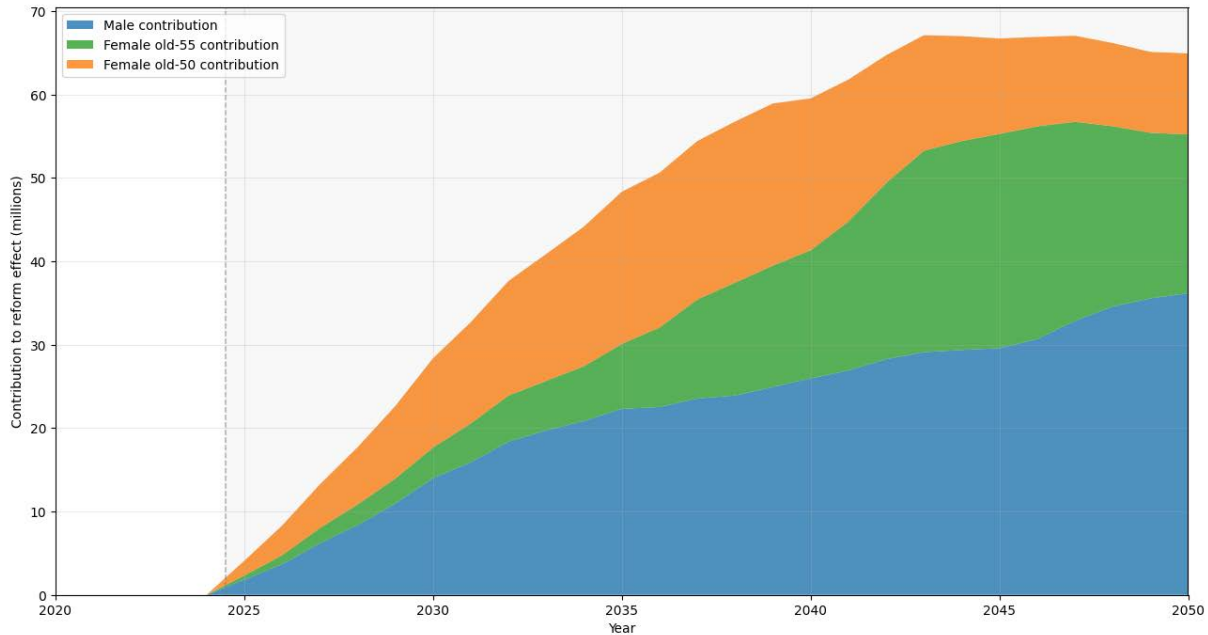
The largest modeling uncertainty is the female split between the old 55-year and old 50-year retirement tracks. This matters because the reform does not raise both female tracks by the same amount. Women in the old-55 track move to 58, while women in the old-50 track move to 55 (Standing Committee of the National People’s Congress, 2024). Assigning too many women to the old-55 track would understate the reform’s effect because it overestimates the share of women who receive the smaller three-year increase rather than the larger five-year increase.

The urbanization-based specification used the national urbanization rate as a direct proxy for the old-55 share, which assigned roughly 64 percent of women to the old-55 track in the 2020 census year, reflecting increases in urbanization over history and into the projected future. That specification is retained as a legacy high-old-55-share comparison (S1). UN-published Chinese census tabulations—UNdata occupation data for 2010 and 2020, coded under ISCO-08—provide a more direct signal. In 2020, analysis of the occupation data suggests that only about 22 percent of employed women worked in core white-collar occupations at the all-age level with only about 12 percent among women aged 50–54, a near-retirement group that is especially relevant for evaluating the old-50 track. The specific classification of roles into blue- and white-collar occupations is defined in Appendix D. The occupation-grounded scenarios therefore assign more women to the old-50 track, which raises the estimated reform effect (United Nations Statistics Division, 2026; author’s calculations).

This explains the main pattern in Table 4.1. At the 2035 benchmark, the occupation-grounded scenarios produce a retained SWA population at least as large as—and in the static-share cases substantially larger than—that of the legacy urbanization proxy. The same holds true for the peak reform effect. The classification of service-and-sales workers affects the estimate but does not drive the result: in the cohort-trended scenarios, varying their classification shifts the 2035 effect by several million people, but the broader conclusion is unchanged—both female tracks are raised under the reform, and the old-50 track receives the larger statutory increase. Full details on the occupation mapping, the service-and-sales sensitivity parameter, and the cohort-trending assumptions are provided in Appendix D and Appendix F.

The distributional implication is more important than the technical one. In the preferred central scenario, women account for roughly 54 percent of the 2035 reform effect and men for 46 percent. Within the female component, the old-50 track—the empirical majority, receiving the five-year increase—contributes roughly 70 percent, or about 38 percent of the total effect across both genders. As successive cohorts enter the reform windows, male and old-55-track female contributions grow through the 2040s (Figure 5.1).

Figure 5.1. Contributions to the Reform Effect, Preferred Central Scenario (S5)



SOURCE: Author's calculations using data from United Nations, Department of Economic and Social Affairs, Population Division, 2024; United Nations Statistics Division, 2026; and the reform schedule in Standing Committee of the National People's Congress, 2024.

NOTE: Decomposition of the additional population retained below statutory retirement thresholds in the preferred central scenario (S5: cohort-trended occupation model, $\theta = 0.25$, $\beta = 0.5$) into male, female old-55-track, and female old-50-track contributions. In 2035, women account for about 54 percent of the effect; the old-50 track contributes 70.1 percent of the female component, or about 38 percent of the total. These are statutory headcounts, not realized employment or labor-force participation.

The burden of this reform may be perceived as inequitable. The aggregate cushion of this policy depends heavily on extending the statutory working lives of blue-collar women, many of whom likely have lower wages, less formal employment, and more physically demanding work, which concentrates the costs of adjustment on the groups least equipped to absorb them. The phase-in distributes the burden unevenly across cohorts as well: early transition cohorts face retirement threshold delays of just a few months, while later cohorts—men born from 1977, old-55-track women born from 1982, and old-50-track women born from 1985—are exposed to the full increase.

Overall, the model does not estimate whether these retained workers will remain employed during those extra years, but it does identify where the legal adjustment is concentrated. The policy question is whether the reform creates longer effective working lives for this group, or simply delays access to retirement benefits for workers least able to absorb the change.

Chapter 6. What the Model Does Not Show

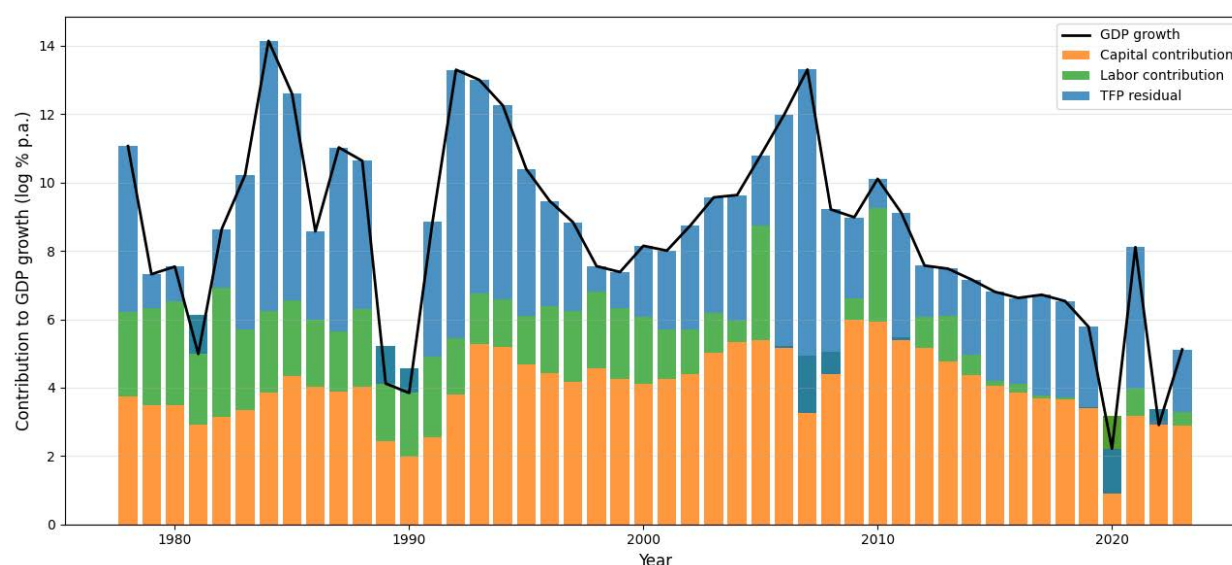
The model is most useful as a policy-timing and headcount simulation by showing the size and distribution of the statutory cushion created by the reform. However, the following features are out of the model's scope:

- **Actual employment or participation.** The model estimates statutory eligibility, not whether people work.
- **Informal work and effective retirement behavior.** Many people work after formal retirement; others exit earlier for health, caregiving, or employment reasons. Many nominally retired women provide childcare and eldercare within families, and formal support systems lean on that unpaid work; retaining these cohorts longer would reshuffle care arrangements as well as labor supply. Age and gender preferences in hiring add a further headwind, and these distributional effects sit outside the headcount.
- **Pension contribution and benefit flows.** The model implies fiscal timing effects but does not quantify receipts, outlays, or contribution compliance.
- **Employer demand for older workers.** Statutory eligibility matters only if firms retain, hire, or retrain older workers.
- **AI's realized labor impact.** The model does not estimate whether AI complements, substitutes for, or displaces the cohorts retained below the threshold.
- **Internal policy intent.** The overlap between the reform window and China's 2035 modernization horizon is analytically important, but the model does not prove intent.

Chapter 7. What the Window Is Worth: The Reform, AI, and 2035

The retirement reform’s timing coincides with China’s 2035 modernization horizon, but how the reform contributes to that goal will also depend on the economy’s productive capacity. The reform’s effect operates through the labor channel — not capital or productivity — but labor is no longer driving Chinese growth. Between 2016 and 2023, labor’s contribution to GDP growth was essentially zero; capital and productivity carried almost all of it (Feenstra, Inklaar, and Timmer, 2015; Figure 7.1). Keeping tens of millions of people in the statutory workforce slows the demographic drag on that contribution, but the reform alone cannot make labor a new engine of growth. Even on the generous assumption that every person the reform retains below the statutory threshold works, the avoided decline in labor input would leave annual growth through 2035 only about 0.3 percentage points higher than under the old rules (Appendix J). The reform buys time; the question is what China does with it.

Figure 7.1. China’s Growth Model Has Narrowed: Contributions to GDP Growth



SOURCE: Author’s calculations using Penn World Table (PWT) 11.0 data (Feenstra, Inklaar, and Timmer, 2015).
NOTE: Annual decomposition of China’s real GDP growth into contributions from capital input, labor input, and total factor productivity, 1978–2023. The labor contribution is labor-input growth weighted by the labor share; TFP is the PWT residual, which the computed decomposition reproduces to within rounding.

Official planning materials indicate how Beijing may try to use this window. An official study volume accompanying the 15th Five-Year Plan proposals—with contributions from People’s Bank of China Governor Pan Gongsheng and Finance Minister Lan Foan—puts the growth required for the 2035 income goal at 4.17 percent a year over the next decade. That

figure follows from a per capita GDP target above \$20,000 and an expected population decline of about 0.2 percent a year (“China Needs 4.17% Annual GDP Growth to Meet 2035 Goal, Publication Shows,” 2025). China’s recent total factor productivity growth has run near 2.2 percent, with labor contributing roughly nothing (Feenstra, Inklaar, and Timmer, 2015; Figure 7.1).² With labor out of the equation, the 4.17 percent requirement relies on both productivity and capital.

Under the 5 percent capital-input scenario, in which capital-input growth slows from its recent pace of about 7.9 percent a year over 2016–2023 to 5 percent (Table 7.1), recent TFP growth of about 2.2 percent would be approximately sufficient to meet the 4.17 percent benchmark. If capital-input growth slows to 3 or 4 percent, required TFP rises to roughly 2.5 to 2.9 percent (Table 7.1; Figure 7.1). However, the contribution from capital is fading, and the weaker it gets, the more the burden shifts to productivity. The reform itself holds the labor base steady, but closing the distance to 4.17 percent depends on productivity rising precisely as the capital engine that has driven Chinese growth loses force. This is why the reform is not a substitute for but instead a complement to the productivity agenda.

Table 7.1. What the 2035 Goal Requires of Productivity as Capital Slows

Capital-input growth	Required TFP growth	Required minus recent TFP growth (pp)
7%	1.23%	–0.96
6%	1.65%	–0.53
5%	2.08%	–0.10
4%	2.51%	0.32
3%	2.94%	0.75

SOURCE: Author’s calculations using PWT 11.0 data (Feenstra, Inklaar, and Timmer, 2015).

NOTE: TFP growth required to meet average annual GDP growth of 4.17 percent through 2035, by assumed capital-input growth. The third column is required TFP growth minus China’s recent TFP growth (the 2016–2023 average of 2.19 percent), in percentage points: a negative value means recent productivity performance would already meet the target, a positive value that it would fall short by that margin. Slower capital growth is the less favorable case because it raises the needed productivity that China must generate to stay on track. Every row assumes capital-input growth below its 2016–2023 average of 7.9 percent. Parameters, the growth-accounting identity, and the full sensitivity logic are in Appendix J.

Whether AI helps close that gap depends on what it does to the people retained below the statutory retirement threshold. The simulation shows that the cushion consists disproportionately of blue-collar women with the old-50 track supplying about 38 percent of the 2035 effect (Chapter 5). The remainder comes from women on the old-55 track and from men who move

² Estimates of China’s recent TFP growth diverge widely; depending on data and method, analysts have judged it extremely low or exceptionally strong (Lee, 2025; “China Is Being Fuelled by Inspiration, Not Perspiration,” 2025). If actual TFP growth has been lower than the PWT series suggests, the gap between recent performance and the 2035 benchmark is wider than Table 7.1 indicates, which reinforces the report’s caution.

from 60 to 63. If AI and automation complement these cohorts by improving matching, easing physical demands, supporting older-worker participation, and expanding labor-intensive services such as elder care, healthcare, logistics, and education, then the statutory window converts into actual employment and output. If AI instead substitutes for tasks prevalent in the occupation groups represented by these cohorts, raising skill requirements faster than older workers can meet them, the reform will have retained precisely the people whose jobs are disappearing. The statutory headcount would still look healthy on paper with tens of millions kept below the retirement threshold, but the number would be hollow. People retained below statutory thresholds but unable to find productive employment would contribute little to output, pensions, or growth. The window would buy time the economy cannot effectively use.

Even if China successfully boosts productivity during this period, its household consumption remains only around 40 percent of GDP, which is significantly lower than that of other major economies.³ The International Monetary Fund (IMF) argues that raising consumption and strengthening social protection are central to sustainable growth (International Monetary Fund, 2026). Without stronger demand, even productive retained workers would have few growing sectors to enter, and excess output the economy cannot absorb adds little to growth. The reform's value through 2035 will be visible in whether the retained cohorts work, contribute, and stay matched, which the indicators in the next section track. To the extent the retained cohorts remain employed, weak domestic demand would push the additional output toward foreign markets, increasing potential trade frictions.

³ By comparison, household consumption is about 60 percent of GDP across OECD members, 68 percent in the United States, 56 percent in Japan, and 50 percent in South Korea (latest available year, 2022–2023). At about 40 percent, China's share is the lowest among major economies and has stayed near that level through two decades of rapid growth (World Bank, undated).

Chapter 8. What to Watch

Whether the reform creates real economic value or merely delays the underlying demographic decline will become observable before 2035. The simulation estimates how many people remain below statutory retirement thresholds, but it does not show whether those cohorts remain healthy enough to work, find employment, or contribute productively. Using census and survey data, Chen and Park (2024) estimate substantial untapped health-related work capacity among older Chinese adults, with greater additional capacity among women than men and among urban than rural residents. This suggests that some older women may have health consistent with working beyond their previous statutory retirement ages. However, the study does not distinguish between women in the old-50 and old-55 retirement tracks, and health-based capacity alone does not imply continued employment, employer demand, or sustained pension contributions. The evidence therefore supports monitoring these outcomes rather than assuming that the larger statutory increase will translate automatically into realized labor supply. Table 8.1 identifies indicators across health and human capital, participation and effective retirement, employment, employer demand, pension contributions, and service-sector absorption that would help determine whether the statutory window becomes productive economic capacity.

Table 8.1. Indicators That Will Show Whether the Window Becomes Capacity

Indicator	What to track	Why it matters
Older-worker participation	Labor-force participation and employment rates for ages 50–64, especially 55–63, by sex (census and labor-yearbook tabulations; effective-retirement surveys such as the China Health and Retirement Longitudinal Study (CHARLS))	Shows whether statutory eligibility translates into actual work
Effective vs. statutory retirement age	The gap between the new statutory retirement ages and the ages at which workers actually exit the labor force, by sex and track (CHARLS; labor-force surveys)	Directly measures whether the statutory window lengthens working lives or remains nominal
Employment of the retained cohorts	Employment and underemployment rates for ages 55–63, separating the old-50 and old-55 tracks, rather than the headline surveyed-unemployment rate	Shows whether the additional people kept below the threshold are absorbed into work or retained only on paper
Women on the old-50 track	Employment, social-insurance contribution continuity, and exit behavior among blue-collar and frontline-service women in the affected cohorts (phasing in from the mid-2020s, reaching the full five-year increase for those born from 1985)	The group receiving the largest statutory increase and carrying most of the reform effect

Indicator	What to track	Why it matters
Pension contributions and benefits	Contribution years and compliance, benefit-claim timing, and fund balances in Ministry of Human Resources and Social Security statistical communiqués	Determines whether the reform actually eases pension pressure
Labor's contribution to growth	Realized labor-input growth and its contribution to GDP (growth-accounting decompositions and PWT updates)	The macro channel the reform operates on; shows whether the retained cushion lifts labor input or the contribution stays near zero
Employer demand for older workers	Vacancy age ceilings, retention, retraining, and re-employment arrangements by sector	Tests whether firms want the cohorts the reform retains
AI: complement or substitute	Occupation-level AI exposure mapped against the retained cohorts' occupational distribution; adoption in labor-intensive manufacturing and services	Connects the retirement window to the broader AI labor-displacement question
Service-sector absorption	Employment growth in elder care, healthcare, education, logistics, and household and business services	Shows whether labor-intensive services expand fast enough to absorb retained capacity
Health and human capital of retained cohorts	Age- and sex-specific functional limitations, self-reported health, physical and cognitive work capacity, and educational attainment by birth cohort, using CHARLS, census, and labor-force data	Tests whether people retained below statutory thresholds can remain employed productively and whether younger, more-educated cohorts alter the distribution of work across ages

SOURCE: Author's analysis.

NOTE: Indicators are observable before 2035 and are grouped from the statutory margin (participation, pension) through the demand-side margin (employer demand, AI exposure, service absorption). Data sources listed are illustrative of where each indicator can be tracked.

Chapter 9. Conclusion

China's 2025 retirement reform is best understood as a temporary statutory buffer. It slows the near-term decline in the population below retirement thresholds, keeping the SWA population steady in the 2025–2035 window. The burden is concentrated on blue-collar women and on younger birth cohorts. It does not reverse aging, and it does not guarantee realized labor supply. It may delay some benefit claims and extend contribution periods, but the model does not quantify pension cash-flow effects.

The policy significance lies in the overlap between that statutory window and China's 2035 modernization horizon, a coincidence that this analysis can quantify but not explain. The reform buys time, but the open question is what Beijing does with it. If AI and automation complement older workers, raise productivity, and expand labor-intensive services, the reform could support an effective demographic transition. If AI instead accelerates skill mismatch, labor-light output, and displacement, it may only postpone a sharper reckoning. The model's contribution is to make the timing visible by showing that a window exists, how large it may be, and which groups bear the adjustment. It does not forecast how China will use the window. That uncertainty is precisely why participation, employer demand, pension behavior, service-sector absorption, and AI's complement-versus-substitute effects should be tracked closely through the mid-2030s.

Appendix A. Data Sources

China's population by sex and single-year age, 1960–2050, is drawn from the UN World Population Prospects 2024 medium-variant, single-age series (United Nations, Department of Economic and Social Affairs, Population Division, 2024). Values through 2023 are treated as WPP historical estimates; values after 2023 are projections, which embed WPP's China-specific mortality, fertility, and migration assumptions under the medium variant. Annual urbanization rates from 1949 are drawn from China's National Bureau of Statistics yearbook series and are used for the legacy S1 specification only (National Bureau of Statistics of China, 2025a). Female occupational shares are drawn from UN-published Chinese census tabulations (2010 and 2020 rounds) under ISCO-08, by sex and five-year age group. UNdata reports that these China occupation records were collected from the census long form and were not estimated up to the total population. The model uses these data to construct occupation shares rather than national employment levels (United Nations Statistics Division, 2026). The reform schedule follows the September 2024 NPC Standing Committee Decision and accompanying State Council implementing measures (Standing Committee of the National People's Congress, 2024).

Appendix B. Mortality

Working-age mortality is not modeled separately. The WPP 2024 single-year age dataset already incorporates WPP's projected Chinese mortality through 2050, so the population at a given age in a given year already incorporates the accumulated mortality under WPP's assumptions (United Nations, Department of Economic and Social Affairs, Population Division, 2024). This inherits the WPP mortality path and avoids introducing a second, potentially inconsistent model. Moreover, the dominant force in the working-age trajectory is cohort entry and exit, not within-cohort attrition/mortality.

Appendix C. The Constructed Measure in Historical Context

The constructed statutory measure is lower than standard age-band aggregates because it applies sex- and occupation-specific thresholds rather than a flat upper age band. The NBS reported 858.0 million people aged 16–59 at year-end 2024, whereas the preferred central scenario’s statutory figure (S5) for 2024 is roughly 781 million (National Bureau of Statistics of China, 2025b). The difference reflects the blue-collar women aged 50–59 (old-50 track) and white-collar women aged 55–59 (old-55 track) who are past their statutory thresholds but inside the 16–59 band. Men do not contribute to this discrepancy because their statutory age coincides with the band’s upper bound. The legacy urbanization-weighted specification (S1) sits higher, reflecting its larger assumed old-55 share.

Appendix D. The Pre-reform Structure and the Female-track Split

Under the rules in effect from 1978 through 2024, men retired at 60, old-55-track women at 55, and old-50-track women at 50, following the conventional cadre/worker (干部/工人) distinction (State Council of the People’s Republic of China, 1978; Standing Committee of the National People’s Congress, 2024). The simulation computes the male statutory population directly and decomposes the female population into the two tracks. The female-track proportional split is the central methodological choice.

The original analysis used each year’s NBS urbanization rate as a proxy for the female old-55 share, with the rural share assigned to the old-50 track and projection years extrapolated from a fitted logistic curve (Appendix E). This specification is retained as S1. Two limitations motivated replacing it as the central case: the proxy is applied to the whole female population rather than to employed women, and the legal cadre/worker distinction is occupational, not residential.

UN-published census tabulations (2010, 2020) report occupation under ISCO-08 by sex and five-year age group (United Nations Statistics Division, 2026). The nine ISCO major groups map onto the cadre/worker distinction as follows, with one ambiguity:

- Old-55 (white-collar): managers, professionals, technicians and associate professionals, and clerical support workers.
- Old-50 (blue-collar): skilled agricultural and forestry workers; craft and related trades; plant and machine operators; elementary occupations.
- Ambiguous: service and sales workers—partly office-adjacent, partly frontline. The data do not disambiguate within the major group, so the assignment is treated as a sensitivity parameter (θ).

The 2020 cross-section shows the urbanization proxy substantially overstates the white-collar share. Despite a roughly 64 percent urbanization rate, the actual share of employed women in core white-collar occupations was about 22 percent at all ages. That white-collar female share fell with age, to roughly 28 percent at 35–39, 17 percent at 45–49, and 12 percent at 50–54 (the cohorts closest to the first female reform windows), and about 6 percent at 55–59. Comparing 2010 and 2020 at fixed cohorts shows women did move into white-collar work over the decade, but well below the pace of urbanization growth, which motivates the sensitivity framework in Appendix F.

Appendix E. The Original Proxy: the Urbanization Model

Why an Urbanization Proxy (S1) was Considered

The legal cadre/worker (干部/工人) distinction that sets the female retirement tracks is not directly observed in the population data: the censuses and the UN single-age series report people by age and sex, not by retirement track. The original analysis therefore needed an observable, time-varying signal that plausibly correlates with the white-collar (old-55) versus blue-collar (old-50) division and is available annually back to 1949 and projectable to 2050. Urbanization was a natural first candidate. China's urban–rural divide tracks the formal/informal and white-collar/manual split more closely than almost any other series the state reports continuously: urban employment skews toward the office, administrative, and professional work associated with the cadre track, while rural employment skews toward agriculture and manual labor associated with the worker track. The urban share is also one of the longest, most complete, and most regularly updated series in Chinese official statistics, which suited a model needing consistent coverage from 1949 through a 2050 projection horizon. The proxy was a defensible first approximation given the data constraints, not an arbitrary choice, which is why it is documented here rather than simply discarded.

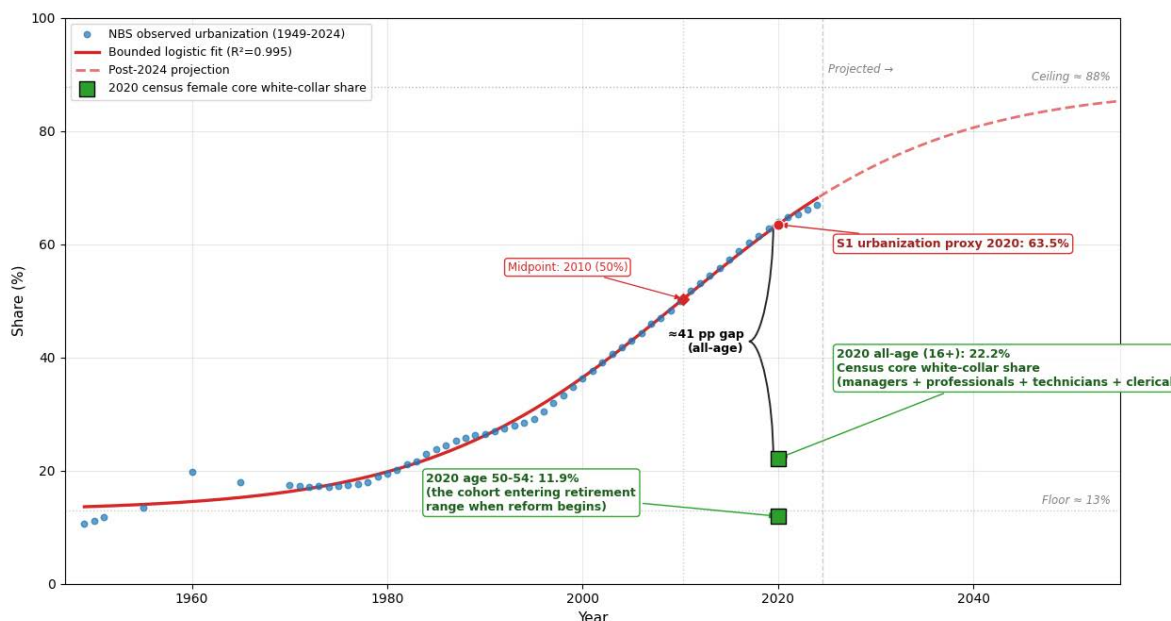
The Bounded Logistic Model

Implementing S1 requires a projection of the urban population. To do this, the urban population share is fitted as a bounded logistic function of the calendar year, using China's National Bureau of Statistics urban-share series for 1949–2024 (National Bureau of Statistics of China, 2025a). A bounded logistic curve is used in preference to logistic regression (generalized linear model): the goal is a smooth S-shaped path for a single percentage series, with the fitted value constrained between an estimated floor and ceiling, both bounded to the 0–100 percent interval. The functional form is as follows:

$$\text{urban share (year)} = \frac{A + (U - A)}{1 + e^{-k(\text{year}) - x_0}} \quad (\text{E.1})$$

where A is the lower asymptote, U the ceiling, k the growth rate, and x_0 the midpoint year. The four parameters are estimated by nonlinear least squares on the 1949–2024 urbanization population share observations. The fit is close (in-sample $R^2 = 0.995$), with $A \approx 12.9$ percent, $U \approx 87.8$ percent, $k \approx 0.076$, and $x_0 \approx 2010$. The fitted urban share is 68.1 percent in 2024 and is projected from the same curve at 69.2 percent in 2025, 74.0 percent in 2030, 77.7 percent in 2035, and rising toward the mid-80s by 2050 (Figure E.1).

Figure E.1. China's Urban Population Share, Observed and Fitted, vs. Census White-Collar Shares



SOURCE: Author's calculations using data from National Bureau of Statistics of China, 2025a; United Nations Statistics Division, 2026 (census occupation shares).

NOTE: Points are NBS observed urban population share, 1949–2024; the solid line is a bounded logistic fit ($R^2 = 0.995$) and the dashed line its projection to 2050. The fitted urban share reaches 68.1 percent in 2024. The census white-collar shares among employed women (about 22 percent at all ages and 12 percent at 50–54 in 2020) are shown for comparison; the gap is why the urbanization proxy (S1) was replaced by the occupation-grounded scenarios as the central specification.

How the Proxy Assigns Retirement Tracks

In the legacy specification (S1), the fitted and projected urban share is used directly as the female old-55 (white-collar) share. The complementary rural share was used as the old-50 (blue-collar) track share. In 2024, S1 thus assigns roughly 32 percent of women to the old-50 track and 68 percent of women to the old-55 track, the latter rising with projected urbanization.

Why was the Proxy Replaced?

The occupation data show this proxy overstates the white-collar share substantially. Urbanization measures where people live, not what they do because many urban residents work in manual, service, or informal jobs that belong to the worker track. The urban share is closer to a potential upper bound on the white-collar share rather than an estimate of it. In 2020, against an urban share of roughly 64 percent, the census shows that only about 22 percent of employed women were in core white-collar occupations at all ages with just 12 percent among women aged 50–54, the cohorts the reform binds first. Because the old-55 track receives the smaller statutory increase (three years versus five years), assigning too many women to it understates the reform's

effect. S1 is therefore retained only as a high-old-55-share comparison; the occupation-grounded scenarios in Appendix F replace it as the basis for the central estimate. Figure E.1 shows the fitted series against the NBS observations and the census white-collar shares, making the gap between the urbanization proxy and the occupation data visible for 2020.

Because the proxy's limitation is one of validity rather than fit, further statistical validation of the urbanization curve, such as out-of-sample testing and holdout projection error, would not change the assessment. Those techniques evaluate how reliably the logistic curve projects urbanization, and the curve fits the urban-share series well ($R^2 = 0.995$). S1's problem is not that it projects urbanization poorly but rather that urbanization is a generally weak proxy that overstates the white-collar share roughly threefold regardless of how accurately the curve is extrapolated. A more rigorously projected urban share would still carry the same specification error. S1 is therefore retained only as a transparent high-old-55-share bound, with the occupation-grounded scenarios (Appendix F) carrying the central estimate. The case for replacing it is rooted in its large discrepancy with the ISCO data that no refinement of the fit could effectively close.

Appendix F. Sensitivity framework: Two parameters, Seven scenarios

The simulations carry two main points of uncertainty. The first is how female service and sales workers divide between the old-55 and old-50 tracks, which no Chinese dataset reports directly (θ). The second is how much of the occupational upgrading observed within birth cohorts between the 2010 and 2020 censuses continues after 2020 (β). This appendix defines the seven scenarios and how they treat these uncertainties.

θ — classification of service and sales workers. θ is the fraction of female service and sales workers assigned to the old-55 track, with the rest assigned to old-50. $\theta = 0$ treats the group as blue-collar; $\theta = 1$ assigns it entirely to the old-55 track. 0.25 and 0.50 are intermediate values, with 0.25 the preferred central value, reflecting the prior that most service and sales work among Chinese women remains informal or frontline (blue-collar) rather than office-adjacent (white-collar).

β — projection of cohort upgrading. β controls how much of the female occupational upgrading (from blue-collar to white-collar) observed between the 2010 and 2020 censuses is projected forward after 2020. The model measures occupational upgrading by tracking birth cohorts across the two censuses. Because a woman aged 30–34 in 2010 is part of the same cohort as a woman aged 40–44 in 2020, comparing that group’s θ -weighted old-55 share at the two dates gives the cohort’s change over the decade. This is a within-cohort trend rather than a comparison of different generations. The model computes this for four such cohort pairs (2010 ages 20–24 through 35–39, matched to 2020 ages 30–34 through 45–49), converts each into an annual rate of change in logit space, and projects each birth cohort forward from its 2020 share at β times its own rate, with the result clipped to a floor of 1 percent and a ceiling of 75 percent. Cohorts born in 1988 or earlier, which would be older than the oldest matched cohort, carry no trend and are held at their 2020 level. $\beta = 0$ freezes the 2020 structure (the static scenarios); $\beta = 0.5$ continues each trend at half its observed rate (applied to all cohort-trended scenarios, S4–S7); $\beta = 1.0$ would fully extrapolate the observed trend but is defined for completeness only. Table F.1 summarizes the seven specifications and their roles in the sensitivity design.

The role labels describe each scenario’s function in the design with no scenario as a definitive upper or lower bound on the reform’s possible effect (Table F.1). The family spans uncertainty in the female-track split, conditional on the model’s structure (population held fixed across the two rule sets, no behavioral response, no participation modeled).

Table F.1. Scenario Definitions

Code	Scenario	θ	β	Role in the sensitivity design
S1	Legacy urbanization proxy (retained for comparison)	—	—	Original specification, retained as a high-old-55-share comparison
S2	Static 2020 occupation shares, all-age	0	0	Cross-sectional anchor
S3	Static 2020 occupation shares, age-specific	0	0	Largest reform effect within the family; freezes the observed 2020 distribution
S4	Cohort-trended occupation model	0	0.5	Preferred range, conservative θ
S5	Cohort-trended occupation model (preferred central)	0.25	0.5	Preferred central specification
S6	Cohort-trended occupation model	0.5	0.5	Additional sensitivity scenario
S7	Cohort-trended occupation model	1	0.5	Upper end of the θ range: all service and sales workers on the old-55 track

SOURCE: Author's assumptions.

NOTE: S2 applies the all-age 2020 share uniformly; S3 holds the 2020 age-specific profile fixed.

Appendix G. The Monthly Cohort Simulation

The simulation estimates the population below the statutory retirement threshold under two sets of rules: the pre-reform system and the 2025 reform. The reform raises retirement ages on a sliding birth-cohort schedule, so each calendar year contains multiple cohorts at different points in the phase-in. Men and women on the old-55 track gain one retirement-age month for every four birth months, capped at 36 months (three years). Women on the old-50 track gain one month for every two birth months, capped at 60 months (five years). The full increase therefore binds for men born in 1977 or later, old-55-track women born in 1982 or later, and old-50-track women born in 1985 or later. The simulation uses single-year UN age data and, within each age year, decomposes the population into twelve monthly sub-cohorts under a uniform-within-year assumption. For each sub-cohort, it infers birth month, applies the cohort-specific delay, and checks status at a mid-year (July) reference point. For each year, the female population is split between blue-collar and white-collar tracks by the scenario share; men are modeled through a single track, with the 60→63 phase-in applied to all.

The old-rules counterfactual is produced by running the same monthly birth-cohort model on the same UN population data, changing only the retirement ages applied, which are the pre-reform ages (60/55/50) instead of the phased-in new ones. Because the two series differ in nothing but the retirement rule, the gap between them isolates the reform's effect.

Appendix H. A Note on θ – β Extrapolation Consistency

The cohort-trended scenarios apply the β trend to the blended θ -weighted old-55 share (Appendix F), rather than projecting the underlying core white-collar and service-and-sales components separately and combining them via θ afterward. Trending the already-blended share is a simplification: it lets θ and the projected trend interact in a way that component-level projection would avoid. This blended approach is irrelevant in the static scenarios (S2, S3) because the 2020 distribution is applied to all future years. Future work could project the two components separately and combine them via θ , isolating the classification choice from the trend.

Appendix I. Reading the Scenario Results

The scenarios remain close in the near term because only a limited number of cohorts have entered the phase-in. They diverge in the late 2030s and 2040s as more cohorts reach the affected retirement ages and the alternative female-track assumptions accumulate. Three points are worth flagging. First, the pre-reform baseline itself varies across scenarios. Because the old retirement ages themselves differ by track (55 for the old-55 track, 50 for the old-50 track), the count of women below the old threshold depends on how women are split between tracks—the same assumption that varies across scenarios (θ and β in the occupation-grounded scenarios, the urban share in S1). The occupation-grounded scenarios put most women on the old-50 track, whose pre-reform age (50) is five years lower, which lowers their old-rules baseline accordingly. Each scenario therefore has its own matched pair of old-rules and reform series, and the reform effect is the gap between them within that scenario—not the gap between a scenario’s reform line and some common baseline.

Second, S3 produces the largest reform effect in the family (about +78 million at 2040, peak +79 million) largely because threshold-age cohorts carry the lowest old-55 shares under its frozen age profile. Third, the scenarios do not converge to a single 2050 value: the reform still holds roughly 64 to 71 million additional people below statutory thresholds in 2050, ranging from 63.6 million in the legacy proxy (S1) to approximately 71.2–71.4 million in the static-share scenarios (S2, S3), with the cohort-trended scenarios (S4–S7) between about 64.7 and 65.5 million. Within the cohort-trended specification, varying θ from 0 to 1 moves the 2035 effect only from 50.5 to 45.6 million, because shifting women between tracks changes the size of each delay (+3 versus +5 years), not whether they are delayed. Because men do not experience an occupational split under the reform, the aggregate men’s +3-year delay results in the largest single block within the three-way decomposition.

Appendix J. Growth Accounting and the 2035 Productivity Benchmark

Chapter 7 places the reform against China's 2035 income goal using a standard growth-accounting decomposition. This part of the appendix sets out the data, parameters, identity, and sensitivity logic behind the figures cited there.

Data and Parameters

The decomposition uses PWT version 11.0 (Feenstra, Inklaar, and Timmer, 2015). Output, capital input, and labor input are taken from PWT's real (inflation-adjusted) series; purchasing-power-parity conversions are not used, since the exercise concerns growth within China rather than cross-country level comparisons. Total factor productivity is not measured directly as it is the residual in the growth-accounting identity, the part of output growth not explained by capital and labor. PWT publishes its own TFP series (rtfpna). Moreover, computing the residual independently from PWT's output, capital, and labor figures reproduces a TFP that is essentially equal to the published TFP series. This confirms internal consistency within the PWT data. Period averages are taken over the following three intervals: 2000–2007, 2008–2015, and 2016–2023. These period intervals are used to show the macro trajectory. The capital share used in the recent-period accounting is $\alpha = 0.427$ (the 2016–2023 average from PWT labor-share data), so the labor share is $1 - \alpha = 0.573$.

Note: PWT's TFP series is one estimate among several, and measures of China's recent TFP growth diverge widely across sources and methods (Lee, 2025). The labor-contribution bound in this appendix is unaffected by that dispute: it is computed from the labor share and the difference in labor-input growth, not from the TFP residual.

The Growth-accounting Identity

Real output growth is decomposed as

$$g_Y = \alpha g_K + (1 - \alpha)g_L + g_A \quad (\text{J.1})$$

where g_Y is real GDP growth, g_K capital-input growth, g_L labor-input growth, α the capital share, and g_A the total factor productivity residual. Rearranged to solve for the productivity terms,

$$g_A = g_Y - (1 - \alpha)g_L - \alpha g_K \quad (\text{J.2})$$

Over 2016–2023 the PWT averages are $g_Y = 5.50$, $g_K = 7.89$, $g_L = -0.08$, and $g_A = 2.19$ percent a year, with $\alpha = 0.427$.

The Narrowing of the Growth Model

TFP growth has slowed across the three periods: about 4.18 percent a year in 2000–2007, 2.43 percent in 2008–2015, and 2.19 percent in 2016–2023. Labor’s contribution to growth is labor-input growth weighted by the labor share. Over the same intervals it fell from about 1.0 percentage point in the 2000s to roughly zero, turning slightly negative in the most recent period: with $g_L = -0.08$ percent and a labor share of 0.573, the contribution is $(1 - \alpha) \cdot g_L \approx -0.05$ percentage points. The statement in Chapter 7 that labor contributes ‘roughly nothing’ refers to the contribution, not to labor-input growth itself. Capital’s contribution also fell from about 5.1 percentage points in 2008–2015 to about 3.4 points in 2016–2023 but remains the largest single component.

The 2035 Benchmark

The productivity target in Chapter 7 is Beijing’s own. An official study volume accompanying the 15th Five-Year Plan proposals, with contributions from People’s Bank of China Governor Pan Gongsheng and Finance Minister Lan Foan, puts the average annual GDP growth required to reach the 2035 income goal at 4.17 percent over the next decade. The figure follows from a per capita GDP target above US\$20,000 by 2035 and an expected population decline of about 0.2 percent a year (“China Needs 4.17% Annual GDP Growth to Meet 2035 Goal, Publication Shows,” 2025). It is a benchmark the analysis tests against, not a finding of this analysis.

Required-TFP Sensitivity

Applying the identity with the target $g_Y = 4.17$ percent, the capital share $\alpha = 0.427$, and labor input held at its 2016–2023 average ($g_L = -0.08$ percent), the TFP growth required to meet the target depends on assumed capital-input growth. The required values are 2.08 percent at 5 percent capital-input growth, 2.51 percent at 4 percent, and 2.94 percent at 3 percent. At the higher capital-input rates of 6 and 7 percent, required TFP falls to 1.65 and 1.23 percent. Recent TFP growth of 2.19 percent therefore just meets the target if capital-input growth holds near 5 percent and falls short as capital-input growth slows below it. For reference, capital-input growth averaged 7.89 percent over 2016–2023, so every row of the sensitivity table assumes a slowdown from the recent pace (Table 7.1). These are illustrative sensitivities, not forecasts; they hold labor input and the capital share fixed and vary only capital-input growth.

The One-for-one Bound on the Reform’s Growth Contribution

Chapter 7 states that, even if every person the reform retains below the statutory threshold were employed, the reform would add only about 0.3 percentage points a year to GDP growth through 2035, by slowing the decline in labor input that would otherwise drag on growth. The bound is constructed as follows. In the preferred central scenario (S5), the SWA base grows at 0.06 percent a year over 2024–2035 under the reform, against -0.51 percent a year under the old rules, which is a difference of about 0.57 percentage points a year in the growth rate of the statutory base. Treating that base one-for-one as labor input (the generous assumption that every retained person works) and weighting by the labor share, the effect on GDP growth is approximately $0.573 \times 0.57 \approx 0.33$ percentage points a year. The “one-for-one” framing is deliberate and generous: PWT labor input embeds hours and human capital, so mapping a statutory headcount onto labor input overstates the true labor contribution. Therefore, the 0.3-point figure is an upper bound, not a central estimate.

Appendix K. Technical Caveats

The policy-level limitations are set out in Chapter 6; the modeling-technical caveats are as follows:

Scenario and Demographic Model (Appendices D-I)

- Mortality is inherited from UN WPP and not separately parameterized, which means the model inherits WPP's own uncertainty.
- Uniform-within-year monthly assumption: population is assumed evenly distributed across the twelve birth months within each age year.
- Female-track classification is a model choice: even with census grounding, the cadre/worker boundary is assumed as a two-group simplification, and the service-and-sales group is handled as a sensitivity parameter precisely because the mapping onto blue-collar and white-collar tracks has uncertainty.
- Cohort upgrading is observed over one decade only: the 2010–2020 shift is the sole cohort observation; β surfaces but does not resolve this uncertainty. The static scenarios act as a no-extrapolation benchmark.
- No behavioral response to the reform: the population is held unchanged across rule sets; effective retirement behavior, take-up of voluntary-delay provisions, and disability flows are not modeled.
- No quantified AI–labor offset: the analysis does not quantify what fraction of the declining statutory population AI would substitute for or augment.

Growth Accounting (Chapter 7 and Appendix J)

- The one-for-one bound on the reform's growth contribution is an upper bound, not an estimate: PWT labor input embeds hours and human capital, so treating a statutory headcount as labor input one-for-one overstates the true labor contribution. The figure of about 0.3 percentage points a year is therefore generous by construction.
- The required-TFP sensitivity is one-dimensional: it holds labor-input growth and the capital share fixed and varies only capital-input growth. It is an illustration of how the productivity requirement moves with investment, not a joint projection. The capital share and labor input could themselves change.
- Penn World Table measurement uncertainty is inherited: the TFP residual, capital stock, and labor-share series carry PWT's construction assumptions and periodic revisions, which the analysis adopts without independent adjustment. This is analogous to the inherited-WPP caveat above.
- The 4.17 percent benchmark is taken as given, not validated as it is Beijing's own arithmetic (per capita GDP above US\$20,000 by 2035, population declining about 0.2 percent a year). This parameter is what the analysis tests against rather than an independent derivation of it.

- TFP is a residual, not a measured quantity: it is what remains after capital and labor are accounted for, so it absorbs measurement error and everything unmodeled, such as allocative efficiency, capacity utilization, and technical change. “Productivity” is a residual label, not a direct observation.

Abbreviations

TFP	total factor productivity
GDP	gross domestic product
SWA	statutory working-age
PWT	Penn World Table
NBS	National Bureau of Statistics (China)
IMF	International Monetary Fund
AI	artificial intelligence
WPP	World Population Prospects
UN	United Nations
ISCO-08	International Standard Classification of Occupations, 2008 revision
CHARLS	China Health and Retirement Longitudinal Study

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About the Author

Sichen (Shawn) Chao is a researcher at RAND. His work examines China's political economy, technology policy, labor-market change, and the strategic implications of artificial intelligence. His current research focuses on AI-induced biorisk, U.S.-China tech competition, China's 2035 modernization agenda, demographic change, and the labor-market consequences of automation and productivity-enhancing technologies.

Chao previously worked on technology and science-and-technology policy issues across the public and private sectors. At the Institute for Defense Analyses, he advised the White House Office of Science and Technology Policy on climate and U.S.-China AI competition. His prior technology experience in Silicon Valley involved working on automating financial tools and building machine-learning models. He holds degrees in mathematics and business analytics from the Massachusetts Institute of Technology (MIT).